

JYOTHI RATNAM

Java Full-Stack Developer | Backend Engineer

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PROFESSIONAL SUMMARY

Java Full-Stack Developer / Backend Engineer with **3.2+ years of experience** building production-grade backend systems, REST APIs, distributed processing pipelines, and real-time applications using **Java and Spring Boot**. Experienced with **Apache Kafka, MQTT, MySQL, React**, and production deployments across **Linux, Tomcat, and NGINX** environments.

TECHNICAL SKILLS

Programming Languages:	Java, SQL, JavaScript
Backend Engineering:	Spring Boot, Spring, Servlets, JSP, REST APIs, Microservices, JPA / Hibernate
Messaging & Streaming:	Apache Kafka, MQTT
Security & Auth:	Spring Security, JWT Authentication, JKS Certificates, API Key Authentication
Databases & Caching:	MySQL, MongoDB, PostgreSQL, Redis
Frontend Development:	React, HTML, CSS, JavaScript
Tools & Infrastructure:	Git, Maven, Docker, NGINX, Apache Tomcat, Postman, Linux
Architecture & Concepts:	Distributed Systems, Event-Driven Architecture, Multithreading, Fault Tolerance, API Design
File Processing:	CSV, XLS, XLSX, Apache POI

PROFESSIONAL EXPERIENCE

Linkwell Telesystems Pvt Ltd — Associate Engineer 19/06/2023 – Present | Hyderabad

- Engineered and maintained scalable backend systems and high-throughput REST APIs using **Java** and **Spring Boot** for business-critical production applications.
- Engineered non-blocking asynchronous processing workflows, optimizing request response times and server resource utilization under heavy traffic.
- Implemented advanced MySQL database indexing and query optimization strategies, substantially reducing database query execution latency.
- Diagnosed and resolved complex application, API, database, and live production server issues to sustain high availability and system reliability.
- Managed full-lifecycle server deployments and automated backend maintenance across Apache Tomcat, NGINX, and Linux server environments.

FEATURED PROJECTS

Corporate Meeting Room Booking System Live Demo | Demo Video: [GitHub](#) | [Portfolio](#)

Tech Stack: Java, Spring Boot, React, REST APIs, MySQL, Spring Security

- Designed and implemented a full-stack corporate meeting reservation application utilizing a **Spring Boot REST API** backend and a **React** single-page frontend.
- Integrated secure user authentication, role-based access control, room availability validation, and booking availability validation and conflict prevention for booking workflows.
- Developed core features including "My Bookings" management, instant meeting cancellations, dynamic QR code verification, and multi-format attachment handling.
- Deployed the React frontend and Spring Boot backend using cloud-hosted infrastructure with MySQL persistence.

File Processing Pipeline using Apache Kafka File Processing Services

Tech Stack: Java, Spring Boot, Apache Kafka, REST APIs, MySQL, Docker, Consumer Groups, DLQ, Linux

- Lowered file processing latency by **40%** through asynchronous Kafka messaging, topic partitioning, and parallel consumer group execution.
- Achieved **99.9% processing reliability** by implementing automated retry handling and Dead Letter Queue (DLQ) exception routing.
- Enabled scalable concurrent data ingestion and processing workflows using multi-partitioned Kafka topics and worker pools.
- Reduced deployment overhead and environment configuration effort by **50%** using Docker containerization.

Tech Stack: Java, JSP, Servlets, REST APIs, MySQL, Apache Tomcat, Apache POI, Linux, CSV/XLS/XLSX

- Managed and optimized enterprise production backend services handling **500k+ records** in high-load server environments.
- Reduced operational human errors by **~60%** by automating multi-stage data validation, bulk upload, and report generation workflows.
- Improved backend data processing efficiency by **45%** through optimized service integration and SQL query execution.
- Sustained **99.9% system uptime** and cut manual operational effort by **70%** by engineering bulk file import/export capabilities (Apache POI).

MQTT Device Communication System**Remote Device Monitoring System****Tech Stack: Java, MQTT, REST APIs, Multithreading, Linux, TCP/IP Communication**

- Implemented real-time bidirectional communication connecting backend services with thousands of concurrent remote hardware devices.
- Engineered scalable topic-based MQTT messaging with configured QoS policies and persistent sessions, achieving **99% message delivery reliability**.
- Reduced device response latency by **30%** through non-blocking multi-threaded TCP/IP socket handlers and asynchronous packet routing.

AppSign — APK Signing Platform**Automated Signing Service****Tech Stack: Java, Spring Boot, REST APIs, JKS Certificates, File Handling, Apache Tomcat, Linux**

- Streamlined Android APK signing operations, reducing manual signing effort by **80%** via automated build pipeline integration.
- Accelerated upload-to-signing turnaround time by **50%** through automated binary validation, cryptographic signing, and release packaging.
- Strengthened mobile application release security using controlled Java KeyStore (JKS) certificate management and restricted key access validation.

Log Upload Server**Secure File Management****Tech Stack: Java Servlets, REST APIs, MySQL, Apache Tomcat, API Key Authentication, File Handling, Linux**

- Secured remote device log uploads by implementing API key authentication, request validation, and access control policies.
- Eliminated unauthorized file access and reduced manual file handling errors by **65%** through structured storage organization.
- Improved log search and diagnostic retrieval efficiency by **40%** with user-isolated file storage architecture and database cataloging.

FPMS — Scanner Analysis Dashboard & Metrics System**Analytics & Diagnostics****Tech Stack: Java, JSP, Servlets, MySQL, Apache Tomcat, SQL Query Optimization, HTML, CSS, JavaScript**

- Delivered a real-time analytics monitoring dashboard for tracking scanner device performance metrics and operational diagnostics.
- Reduced SQL query execution time by **35%** through database index optimization and query execution plan tuning.
- Improved operational monitoring efficiency by **50%** with dynamic backend-driven reporting and real-time metrics visualization.

EDUCATION